Measuring AI-Assisted Academic Search & Comprehension in UNC Library Search

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Current Landscape

UNC Library Search (Articles+) provides access to thousands of academic sources but:

- We only track surface-level engagement (clicks, log-ins, downloads)
- We do not measure:
  - How well students understand articles
  - Whether they can choose high-quality sources
  - Where they get stuck in the search interface

Problem Context

Why It Matters

Students struggle with:

- Crafting effective search queries
- Understanding abstracts/methods
- Knowing which articles are empirical, peer-reviewed, or relevant

Libraries struggle with:

- Identifying help needs
- Supporting search literacy
- Ensuring comprehension, not just access

Test whether AI assistance (ChatGPT) improves speed, accuracy, & comprehension during an academic search task.

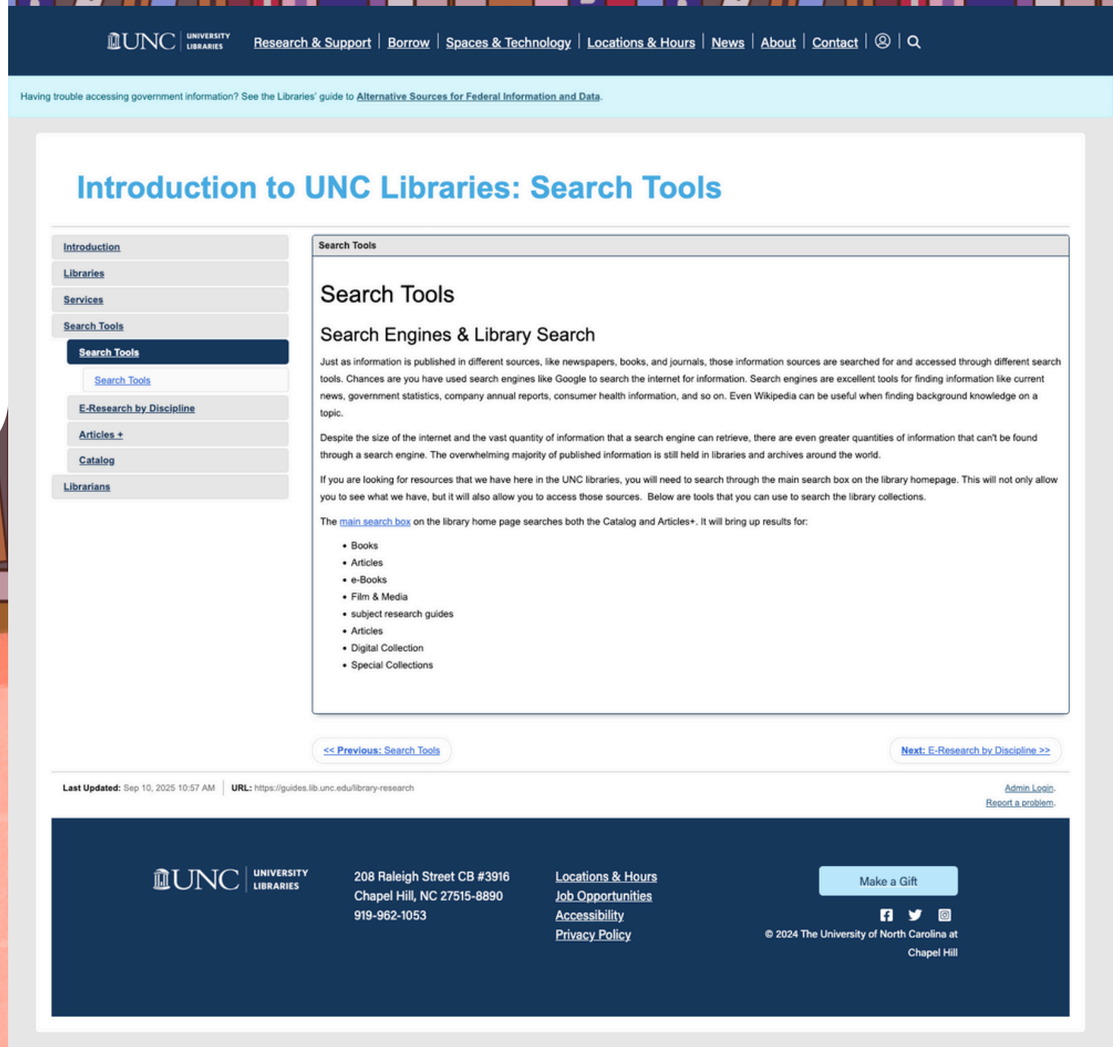
UNC Library Search

Articles+

Searches digital databases and journal repositories, aggregating scholarly literature from multiple fields with advanced filtering.

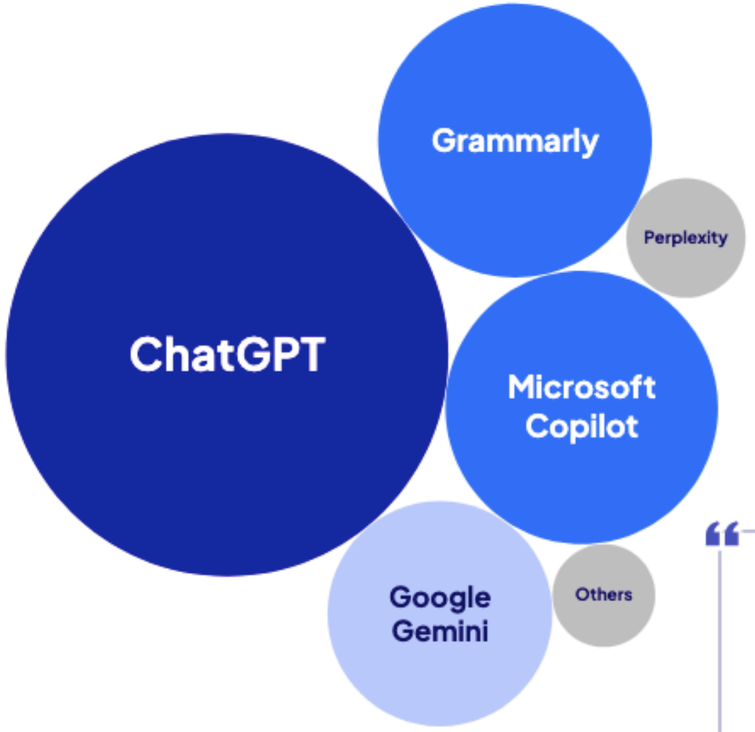
The Catalog

Indexes both physical and digital collections, providing a comprehensive overview of UNC's holdings.



Most used AI tools by students and number of AI tools used

Question: Which AI tools do you use for your course/programme?



Source: Digital Education Council Global AI Student Survey, 2024

Why We Selected ChatGPT

> The **Digital Education Council Global AI Student Survey (2024)** found that 66% of students use ChatGPT, primarily for "searching for information" and "summarizing documents." However, 80% reported that institutional AI integration did not fully meet expectations.

> The **Campbell University AI Tool Usage Survey (2025)** reported AI chatbots (ChatGPT, Microsoft Copilot) as the most utilized tools (63%) among students and faculty, followed by AI writing assistants (48.1%).

> A meta-summary (Campbell Academic Technology Services, 2025b) concluded that AI use is shifting from productivity-oriented applications toward tools that support information seeking, comprehension, and metacognitive learning.

Future work: Implement testing on other highly used AI toold for search literacy (ie. Perplexity Google Gemini)

AI Tool + Platform Analysis

Tool	Purpose	Strengths	Limitations
UNC Library Search (Articles+)	Institutional academic search	Accurate metadata, filters, peer-reviewed sources	Hard for novices; confusing filters; doesn't measure comprehension
ChatGPT	Search coaching, summarization, self-check	Improves query clarity; explains abstracts; supports reasoning	May hallucinate citations; quality depends on prompt clarity

Research Questions

How does ChatGPT assistance influence students' search efficiency, comprehension, cognitive load, and reasoning quality during academic search tasks?

- Does AI support help students locate relevant articles faster or more accurately?
- Does it deepen comprehension of research relevance?
- Does it reduce cognitive load?
- How does it affect students’ motivation, confidence, and anxiety?
- How do students adaptively prompt and refine queries through AI scaffolding?

Pilot Purpose

Before running with real participants, we tested the full workflow to ensure:

- Instructions were clear
- Timing was realistic
- Metrics were measurable
- Scoring criteria made sense

Task

“Find one peer-reviewed empirical article (2019+) on AI-assisted writing in undergraduate education using UNC Library Search, open the full text, write a 4-sentence takeaway, and export the citation.”

Conditions

Group 1: Library Search only
Group 2: Library Search + ChatGPT Prompt Card

Scenario Walkthrough

Refine Search Queries

Students use ChatGPT to improve keyword selection and filter usage in UNC Library Search.

Interpret Abstracts

ChatGPT assists in understanding complex academic abstracts, helping students grasp core findings quickly.

Verify Relevance

AI helps students verify if an article truly addresses their research question, avoiding irrelevant selections.

Summarize Findings

Students craft 2-3 sentence summaries of articles with AI assistance, demonstrating comprehension.

Student responses (summaries and reflections) will be evaluated against established criteria for accuracy, clarity, and relevance, with a focus on guiding users toward valid scholarly sources.

Figure 1.

Example Search Result from ChatGPT (free model GPT-5)

Help me create a precise search query in the UNC Libraries database to find a peer-reviewed empirical study published after 2019 about AI-assisted writing in undergraduate education. Suggest filters or keywords to use.

To create a precise search query for finding a peer-reviewed empirical study published after 2019 about AI-assisted writing in undergraduate education in the UNC Libraries database, here's a step-by-step approach:

1. Basic Search Query Components

Start by combining keywords that are directly relevant to your topic. Here's a breakdown:

- **Keywords for AI-assisted writing**:
  - "AI-assisted writing"
  - "artificial intelligence in writing"
  - "AI writing tools"
  - "automated writing systems"
- **Keywords for undergraduate education**:
  - "undergraduate education"
  - "college students"
  - "higher education"
- **Empirical study focus**:
  - "empirical study"
  - "quantitative study"
  - "qualitative research"
  - "research study"
  - "case study"
- **Time filter for publications after 2019**:
  - Use "after 2019" or "since 2019" to limit the search to recent studies.

2. Constructing the Query

You can combine the above components with Boolean operators (AND, OR, NOT) to refine your search. Here's an example of how you might structure your search:

Search Query Example:

```arduino
("AI-assisted writing" OR "artificial intelligence in writing" OR "AI writing tools" OR "automated
```

Copy code

3. Filters to Apply

In the UNC Libraries database, you can apply several filters to narrow down your results:

- **Peer-reviewed articles**: Make sure to check the box for peer-reviewed articles.
- **Publication date**: Limit the publication date to **2019-present** to ensure you're getting the most recent studies.
- **Language**: If you need English-language articles, filter by language.
- **Article type**: Select "Research Article" or "Journal Article" to avoid reviews or opinion pieces.

4. Advanced Search Options (if available)

Some library databases have advanced search features that allow you to narrow your search further:

- **Subject**: Look for specific subject areas related to "Writing Instruction," "Educational Technology," or "AI in Education."
- **Full-text**: If you want to access the full text of the articles, ensure this filter is selected.

5. Refining Your Search

Once you have initial results, you can refine further by reviewing abstracts for relevance and narrowing by specific journals or additional keywords like "writing pedagogy" or "AI writing tools in education."

6. Alternative Search Phrasing

If the above query doesn't yield enough results, you can try using alternative phrasing for broader coverage. For example:

- **"Artificial intelligence" AND "writing instruction" AND "undergraduate" AND "empirical study"**
- **"AI in higher education" AND "writing tools" AND "college students" AND "research" AND "since**

ised on your findings, you should be able to locate peer-reviewed empirical studies
d writing in undergraduate education.

What We're Measuring

1. Efficiency / Behavior
   - Time to complete task
   - Query reformulations
   - Navigation behaviors
   - Use of filters
   - For AI condition: number & type of prompts

2. Comprehension

Takeaway Rubric (0–3):
   - What the study did
   - Who the participants were
   - Key findings

3. Engagement / Experience
   - Confidence
   - Perceived difficulty
   - Effort
   - AI helpfulness
   - AI dependence

Control Group

Students conduct library searches unaided, using only the UNC Library Search interface.

Experimental Group

Students use ChatGPT alongside Library Search for query formulation, interpretation, and article selection.

Evaluation Framework

Behavioral Efficiency

Measures observable performance: time-on-task, query reformulations, precision at 'k' results. These indicate how effectively students navigate search systems (Kelly, 2009).

Cognitive Comprehension

Assesses understanding, interpretation, and synthesis of content using Day and Park's (2005) six-level taxonomy, differentiating surface from deeper analytical reasoning.

Reasoning & Critical Thinking

Analyzes the quality of explanations and justifications in article summaries, drawing on Facione's (1990) critical thinking framework and Shang et al. (2025).

Cognitive Load

Utilizes the Differentiated Cognitive Load Questionnaire (DCLQ; Klepsch et al., 2017) to distinguish intrinsic, extraneous, and germane cognitive load during the task, based on Sweller's (1988) theory.

Affective & Motivational Engagement

Captures emotional and motivational aspects through interest, anxiety items, and the System Usability Scale (SUS; Brooke, 1996) for both Library Search and ChatGPT.

Evaluation Framework

AI_evaluation | • CSV edit AI_evaluation

Read options | Write options | Preview

Add row | Add column | Apply changes to file and save | Apply changes to file | Hide comments | Tools

	column 1	column 2	column 3
1	Title	Question Type	Options
2	Cognitive Load Section	SECTION_DESCRIPTION	
3	The search task was complex.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
4	The search task required a great deal of mental effort.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
5	The interface was confusing.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
6	I had to exert effort dealing with unclear parts of the task.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
7	I invested mental effort evaluating article relevance.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
8	I tried to understand how the article related to the research question.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
9	Motivation Section	SECTION_DESCRIPTION	
10	I tried to understand the task deeply.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
11	I made an effort to ensure the article matched the research question.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
12	I found the task engaging.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
13	I felt motivated to complete the task well.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
14	Anxiety Section	SECTION_DESCRIPTION	
15	I felt nervous during this task.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
16	I worried about selecting an incorrect article.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
17	I felt overwhelmed by the number of search results.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
18	System Usability Section (SUS)	SECTION_DESCRIPTION	
19	I think I would like to use this system frequently.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
20	I found the system unnecessarily complex.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
21	I thought the system was easy to use.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
22	I think I would need assistance to use the system.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
23	I found the functions were well integrated.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
24	I thought the system had too much inconsistency.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
25	I imagine most people would learn this system quickly.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
26	I found the system cumbersome to use.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree

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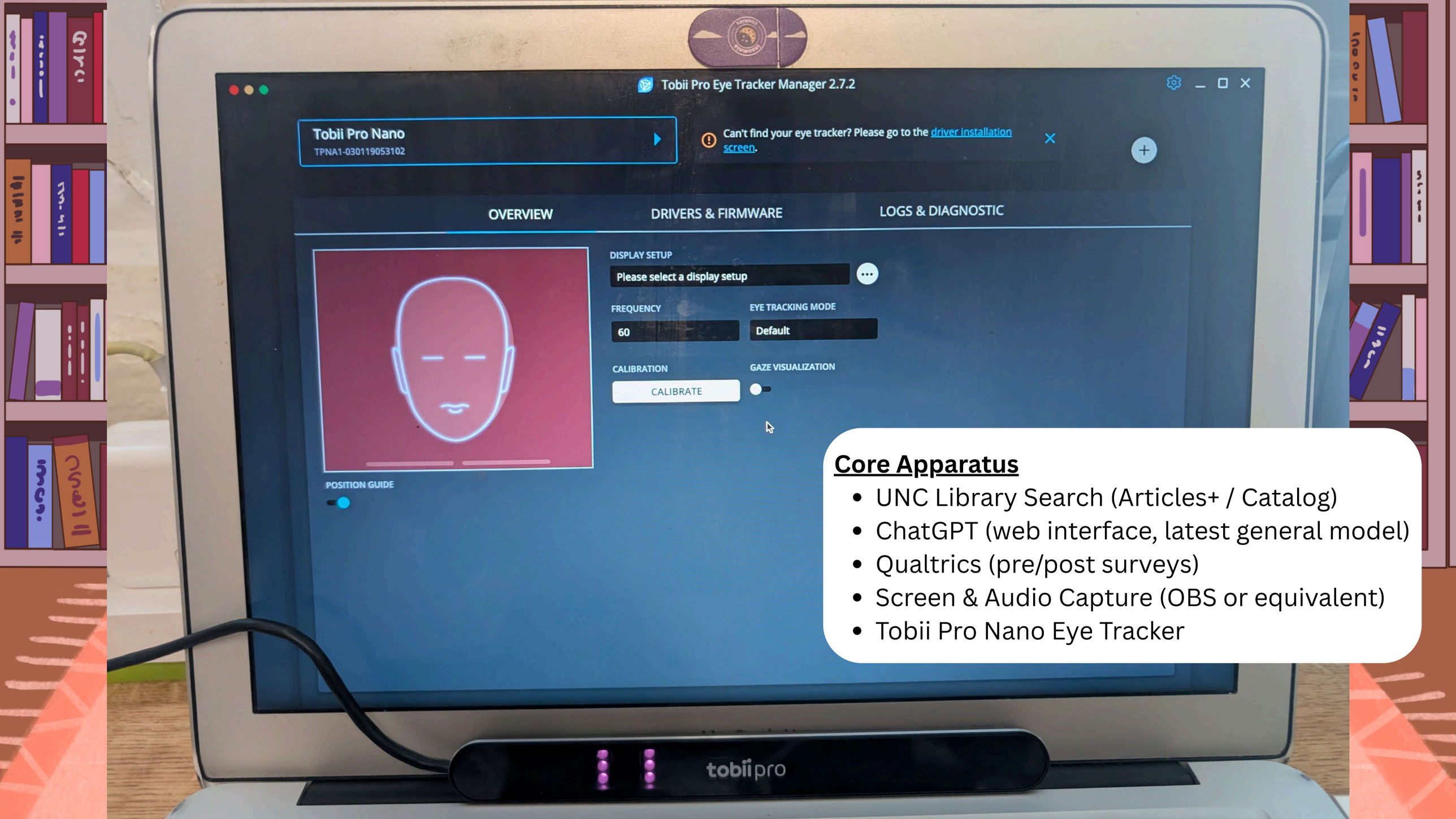
26	I found the system cumbersome to use.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
27	I felt confident using the system.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
28	I needed to learn a lot before I could get going.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Agree;Strongly Agree
29	Comprehension Section	SECTION_DESCRIPTION	
30	Please explain in 2–3 sentences how the article addresses the research question.	PARAGRAPH	
31	How confident are you that the selected article is relevant?	MULTIPLE_CHOICE	Not at all confident;Slightly confident;Moderately confident;
32	ChatGPT Helpfulness Section (AI group only)	SECTION_DESCRIPTION	
33	ChatGPT helped refine queries.	MULTIPLE_CHOICE	Not at all helpful;Slightly helpful;Moderately helpful;Very hel
34	ChatGPT helped interpret abstracts.	MULTIPLE_CHOICE	Not at all helpful;Slightly helpful;Moderately helpful;Very hel
35	ChatGPT helped evaluate article relevance.	MULTIPLE_CHOICE	Not at all helpful;Slightly helpful;Moderately helpful;Very hel
36	Reflection Section	SECTION_DESCRIPTION	
37	What was the most challenging part of this task?	PARAGRAPH	
38	Describe the strategy you used to select your article.	PARAGRAPH	
39	If you used ChatGPT	describe one way it changed your usual search behavior.	PARAGRAPH



Core Apparatus

- UNC Library Search (Articles+ / Catalog)
- ChatGPT (web interface, latest general model)
- Qualtrics (pre/post surveys)
- Screen & Audio Capture (OBS or equivalent)
- Tobii Pro Nano Eye Tracker

Evaluation Framework

Metrics and Methodologies

1. **Behavioral Efficiency**
   Task completion, time on task, query reformulations, use of filters.
2. **Cognitive Comprehension**
   Summary and explanation quality based on Day & Park's six-level taxonomy.
3. **Reasoning & Critical Thinking**
   Quality of explanations and questions using Facione and Shang et al.'s frameworks.
4. **Cognitive Load**
   Intrinsic, Extraneous, and Germane Load measured by DCLQ (Klepsch et al., 2017).
5. **Affective Engagement**
   Motivation, confidence, anxiety, and perceived AI support using scales and SUS.
6. **Eye-Tracking**
   Fixation, saccade, and pupil dilation metrics for objective attention and cognitive load.

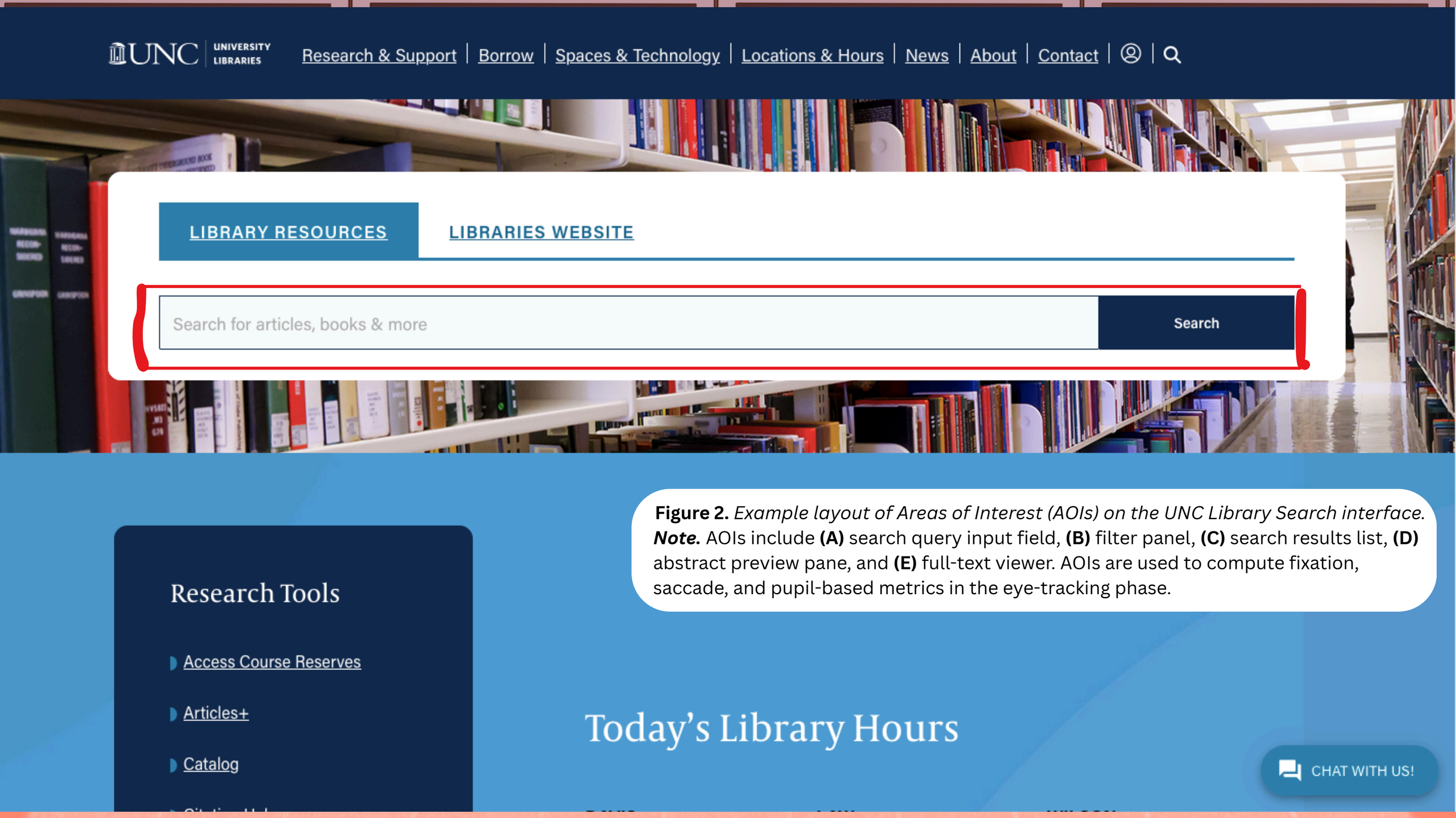


Figure 2. *Example layout of Areas of Interest (AOIs) on the UNC Library Search interface.* ***Note.*** AOIs include **(A)** search query input field, **(B)** filter panel, **(C)** search results list, **(D)** abstract preview pane, and **(E)** full-text viewer. AOIs are used to compute fixation, saccade, and pupil-based metrics in the eye-tracking phase.

Data Analysis

Behavioral Data

Descriptive statistics (M, SD) for time on task, query count, filter usage. Group comparisons using t-tests or Mann–Whitney U tests, and success rates via χ^2 tests.

Comprehension & Reasoning

Rubric-based scoring of summaries and comprehension questions, with inter-rater reliability checks. Group comparisons and correlations with efficiency and cognitive load metrics.

Questionnaire Data

Analysis of DCLQ subscales, motivation, anxiety, and SUS scores. Exploration of relationships between these factors (e.g., SUS vs. ECL).

Process & Think-Aloud Logs

Qualitative coding of verbalizations to identify strategy shifts, confusion, and AI reliance. Prompt log analysis for AI group to classify and evaluate prompt quality.

Eye-Tracking

Integration of Tobii Pro Nano data for fixation, saccade, and pupil metrics. Correlation of gaze data with cognitive load, comprehension, and search efficiency to triangulate findings.

Responsible Use of AI

1. Accuracy & Verification
   - Prompt card explicitly prevents hallucinations
   - Students must still open full text to verify content
2. Transparency
   - Prompts require the AI to explain its reasoning
   - Students rate how much they relied on AI
3. Academic Integrity
   - Takeaway must be written in the student’s own words
   - AI summaries cannot be copied directly
4. Safety & Equity
   - Supports students with varying search literacy
   - Doesn’t replace learning

Design Proposal / Solution Roadmap

Create an AI-supported library search workflow that improves search literacy and comprehension without reducing academic integrity.

Proposed Design Components

1) AI-Guided Search Query Coach/Helper
AI feature that helps students create better search queries.
2) Abstract Explainer Module
2–3 sentence plain-language summary
Highlights study type + population
3) Relevance Checker
Confirms peer-reviewed, empirical, 2019+
Flags mismatches
4) Comprehension Self-Check
2–3 mini questions to reinforce/check for understanding

Impact

Enhanced Search Literacy
Developing new metrics to assess how students effectively utilize search tools and comprehend academic content.

Integrated Research
Integrating AI and eye-tracking technologies to advance research in academic behavior and learning analytics.

Practical Recommendations
Providing actionable insights to UNC Libraries for optimizing digital learning ecosystems.

Holistic AI Evaluation
Offering a comprehensive understanding of AI's role in supporting student learning outcomes.

Extending Prior Work
Building upon foundational research in EFL reading to inform academic search in a library context.

Limitations

- Pilot sample: extremely small (not representative)
- Researcher performing the pilot may be biased
- Task focuses on one search topic
- ChatGPT behavior varies by model version
- Articles+ interface may change over time

Future Research

- Test with real undergraduate participants
- Add optional eye-tracking to measure cognitive load
- Compare multiple AI tools (Copilot, Gemini)
- Expand tasks beyond article search (e.g., synthesis tasks)

Conclusion

- Students struggle with query construction, abstract interpretation, and identifying empirical articles.
- AI when structured, transparent, and responsible has the potential to improve efficiency, accuracy, and comprehension.
- Our pilot shows the workflow, metrics, and evaluation criteria are feasible for the full study.
- The proposed AI-supported design could meaningfully strengthen search literacy across UNC’s learning ecosystem.

THANK YOU